REPUBLIC OF CAMEROON
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MINISTRY OF HIGHER EDUCATION
EDUCATION

REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie
MINISTERE DE L’ENSEIGNEMENT SUPERIEUR

AI-BASED SIGN LANGUAGE RECOGNITION SYSTEM

Presented by:

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Outline

- Introduction
- Problem statement
- Project objectives
- Software Methodology, Tools and Technologies
- System analysis and Design
- Conclusion

Introduction

- Communication is a fundamental human need that enables people to express ideas, emotions, and information in daily life. However, for individuals who are deaf or non-verbal, communication with hearing and verbal people is often difficult because sign language is not widely understood. This communication gap limits their participation in education, healthcare, public services, and social interactions.
- With the advancement of Artificial Intelligence (AI) and deep learning, it is now possible to build systems that can recognize human gestures and translate them into understandable text or speech. This project explores how AI can be used to improve accessibility and promote inclusion by enabling real-time sign language translation using a camera-based system.

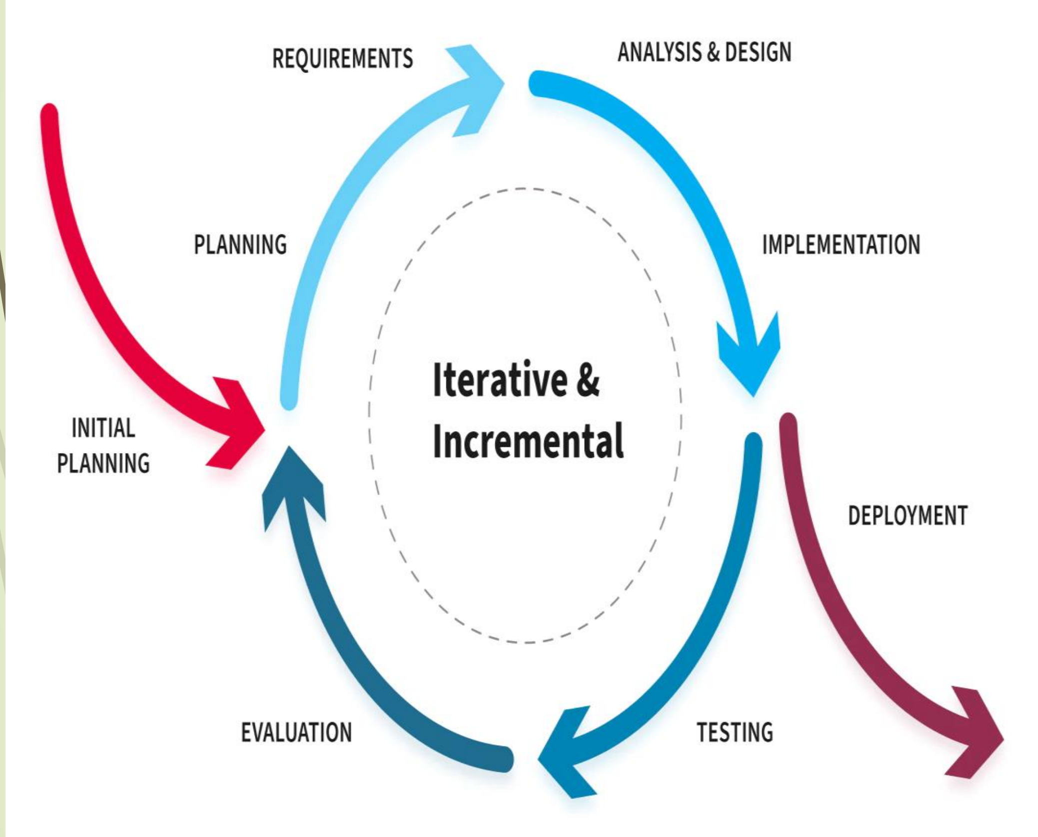
Problem Statement

- Despite the importance of sign language, most people do not understand it, creating a major communication barrier between the deaf and non-verbal community and the general public. There is also a shortage of professional sign language interpreters, especially in public institutions such as hospitals, schools, banks, and government offices.
- As a result, deaf and non-verbal individuals often struggle to communicate their needs effectively, leading to misunderstandings, exclusion, and reduced access to essential services. There is therefore a need for an intelligent, affordable, and automated system that can translate sign language gestures into text or speech in real time, without requiring a human interpreter.

Project Objectives

- **Primary Objective**
  - The main objective of this project is to design and implement an AI-based sign language recognition system that can translate sign language gestures into text and speech to improve communication and accessibility.
- **Secondary Objectives**
  - To capture sign language gestures using a camera.
  - To train a deep learning model capable of recognizing sign language gestures.
  - To translate recognized gestures into readable text and audible speech.
  - To design a web-based interface that allows easy interaction with the system.
  - To enhance inclusion and accessibility for the deaf and verbal-impaired community.

Software Methodology ,Tools and Technologies



- This project adopts the **Agile software development methodology**, which supports iterative and incremental development. Agile allows continuous testing, regular improvements, and flexibility to adapt to changes during development.
- The system is developed using modern tools and technologies.
  - Python is used for AI model development,
  - TensorFlow and Keras for deep learning.
  - OpenCV is used for video and gesture processing.
  - The web interface is built using React and Tailwind CSS,
  - Flask or FastAPI handles backend services.
  - The database is managed using MySQL through MySQL Workbench.

System Analysis and Design

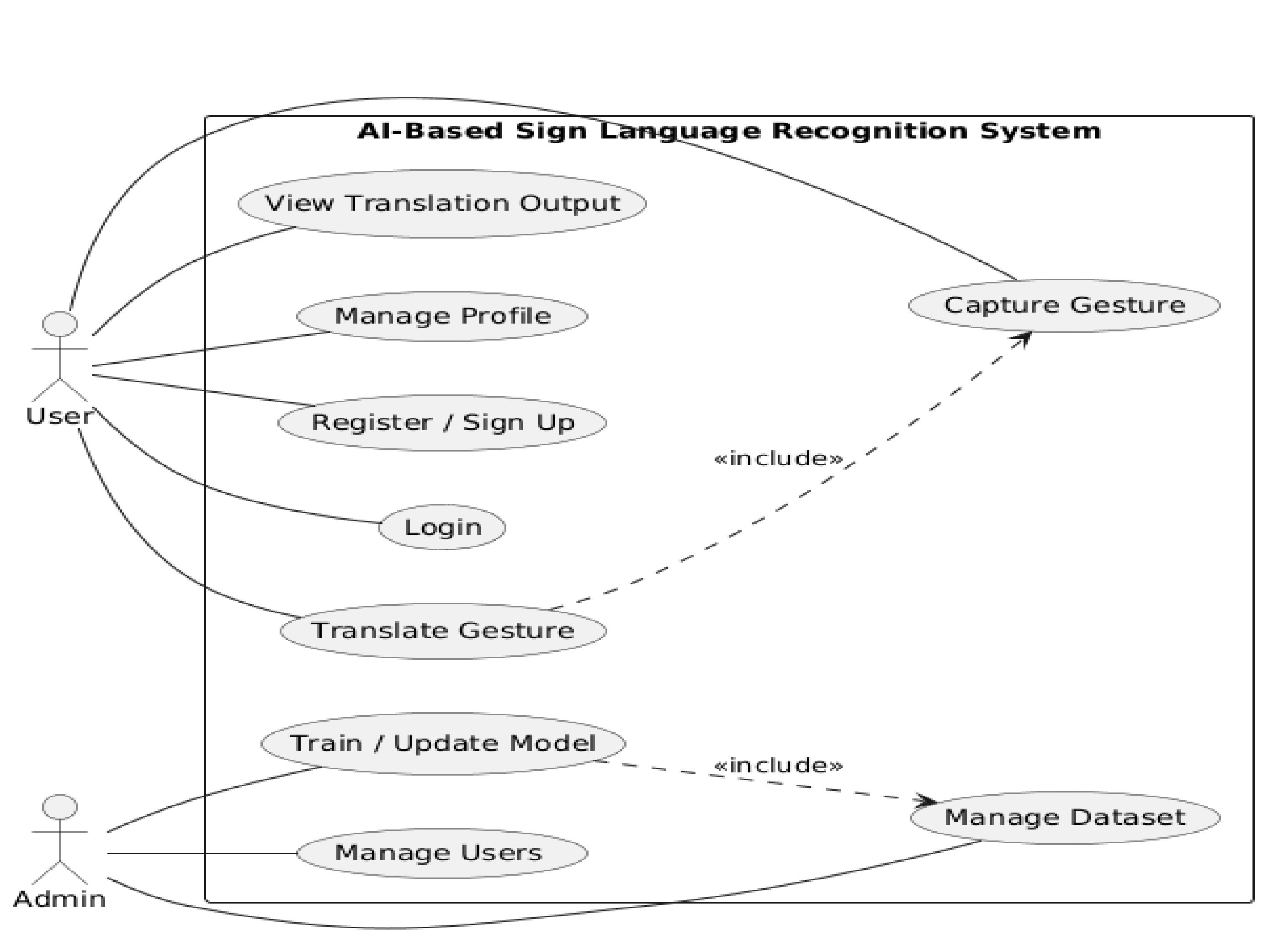
- **Deep Learning Lifecycle for Sign Language Recognition System**
  - **Data Preparation** – Kaggle WLASL sign language dataset
  - **Model Engineering** – CNN–LSTM training on Google Colab
  - **Evaluation & Validation** – Accuracy testing and validation
  - **Deployment** – Web-based AI integration
  - **Monitoring & Maintenance** – Model updates and retraining

Requirement Analysis

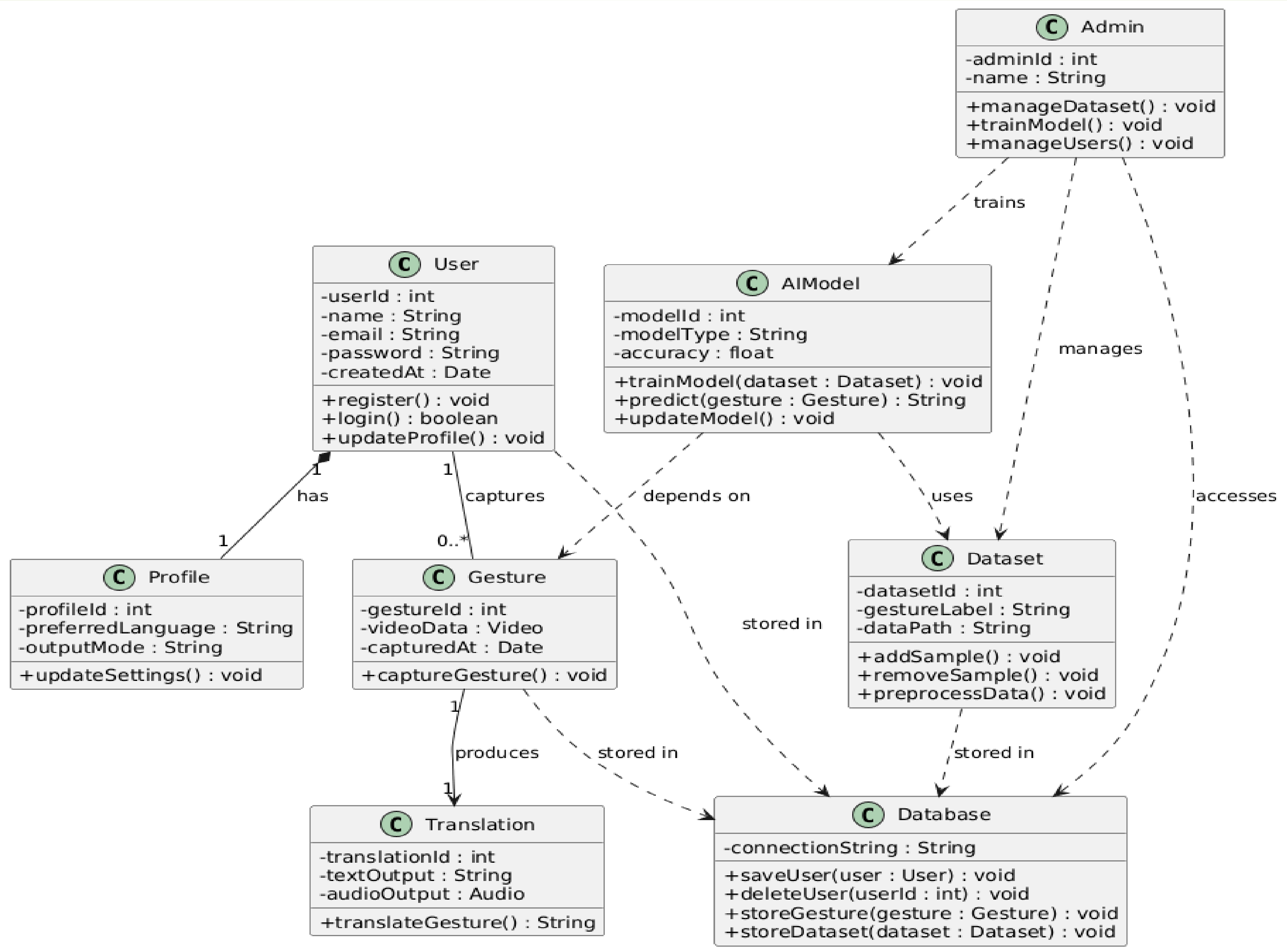
- **Functional Requirements**
  - The system should allow users to sign up and log in,
  - capture gestures using a camera,
  - translate gestures into text and speech,
  - and display translation results in real time.
- **Non-Functional Requirements**
  - The system should be user-friendly, responsive, secure,
  - Capable of processing gestures with minimal delay.
  - It should also be reliable and scalable for future improvements.

System Modelling with UML

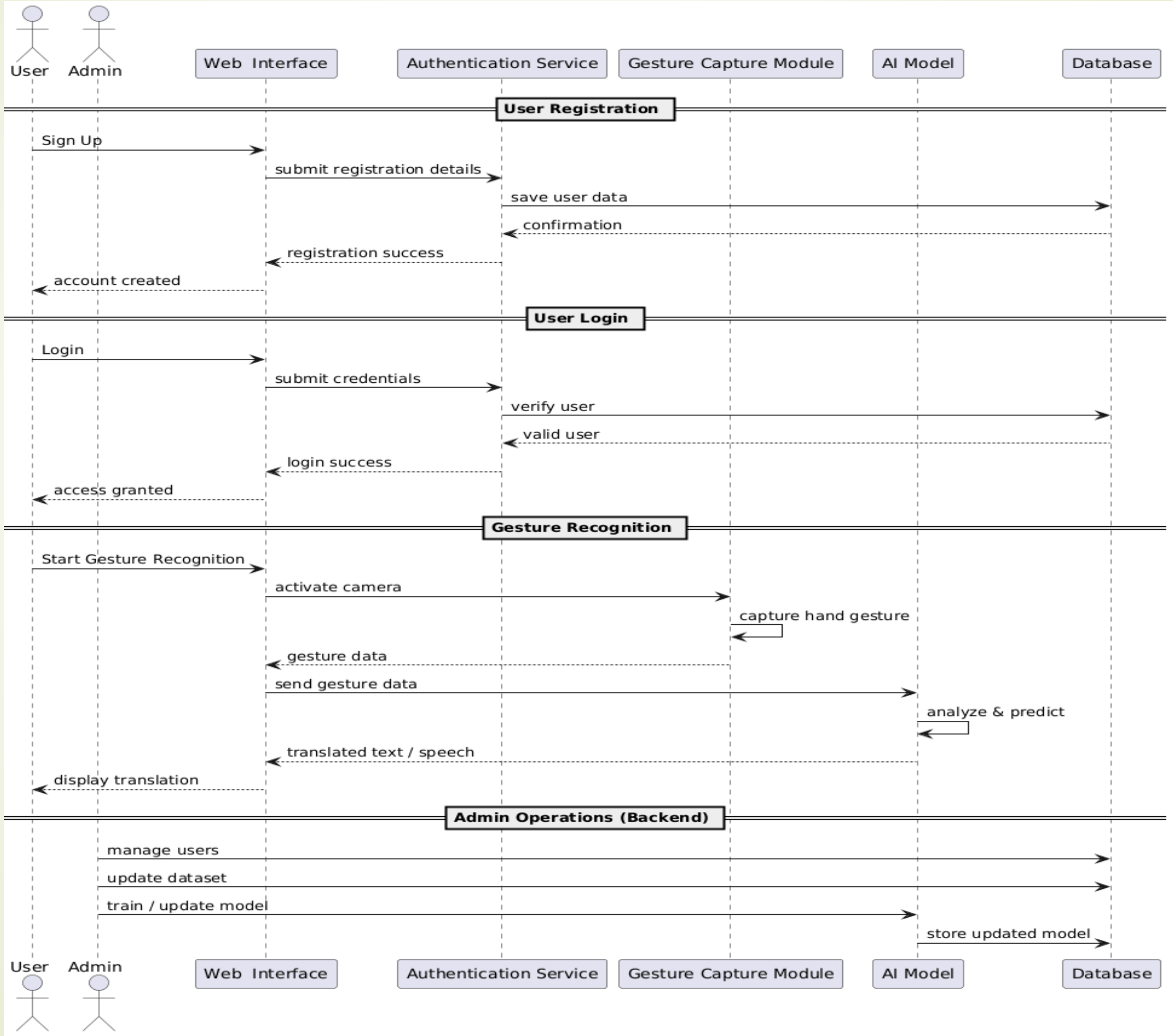
- USE CASE DIAGRAM



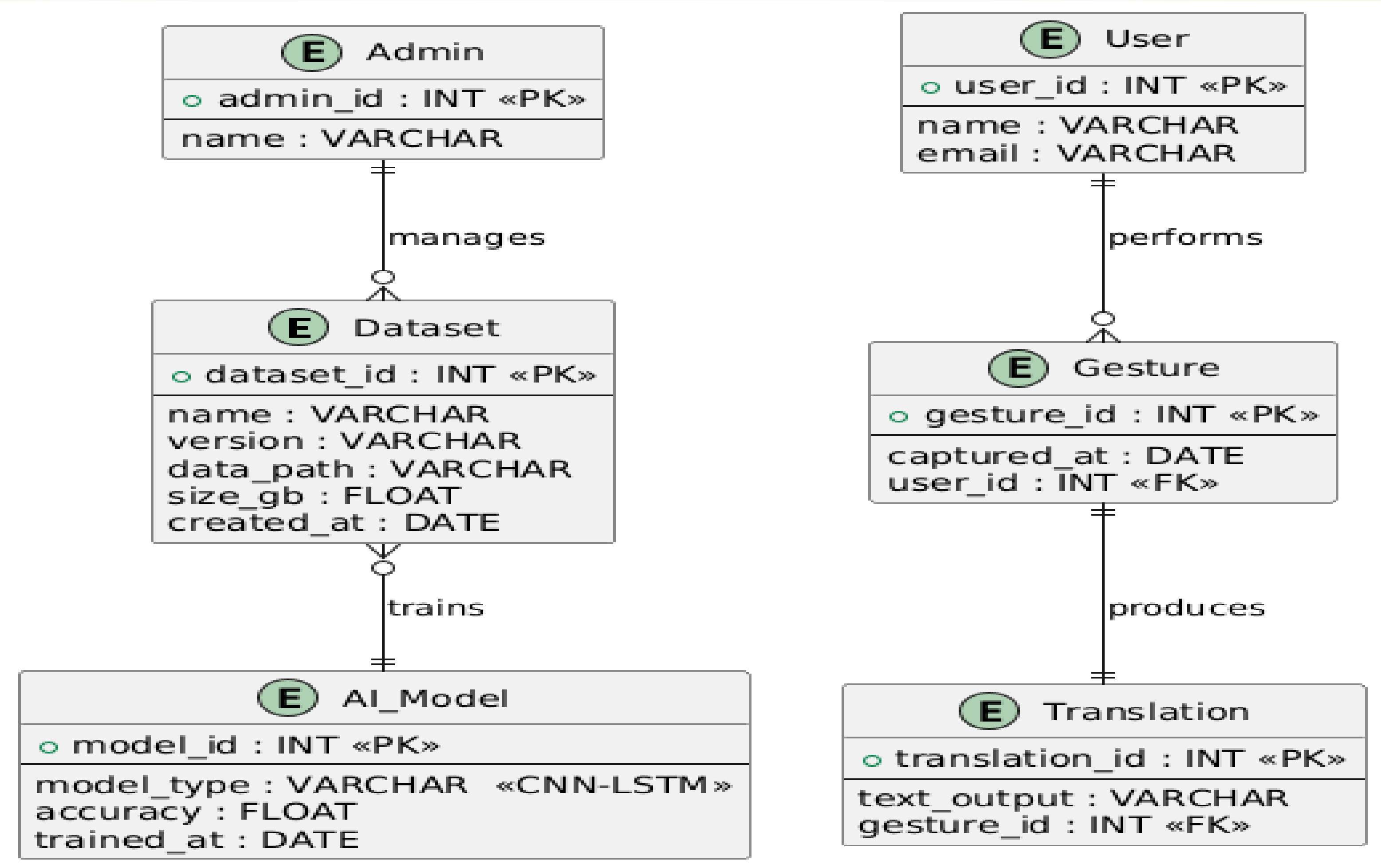
CLASS DIAGRAM



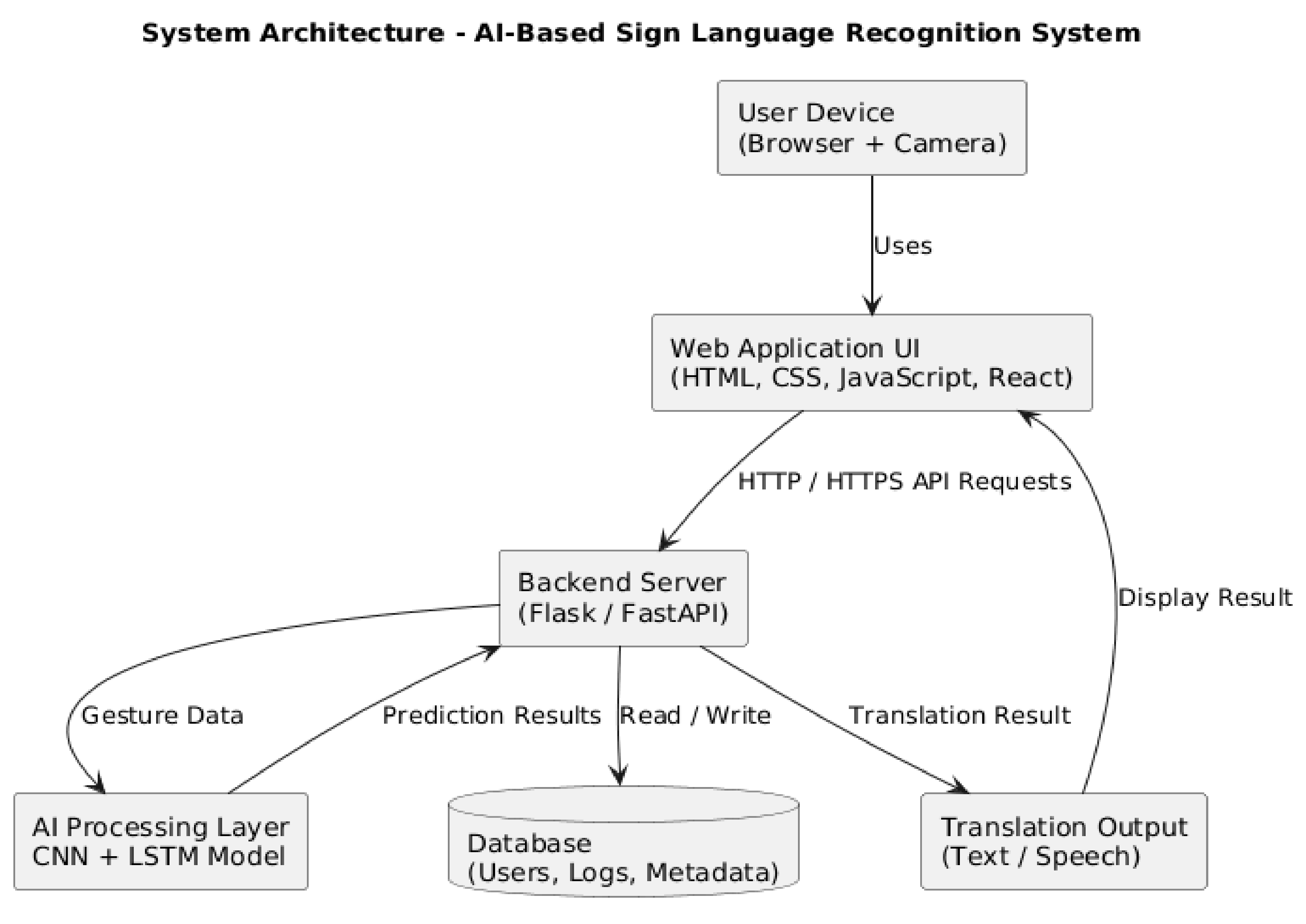
SEQUENCE DIAGRAM



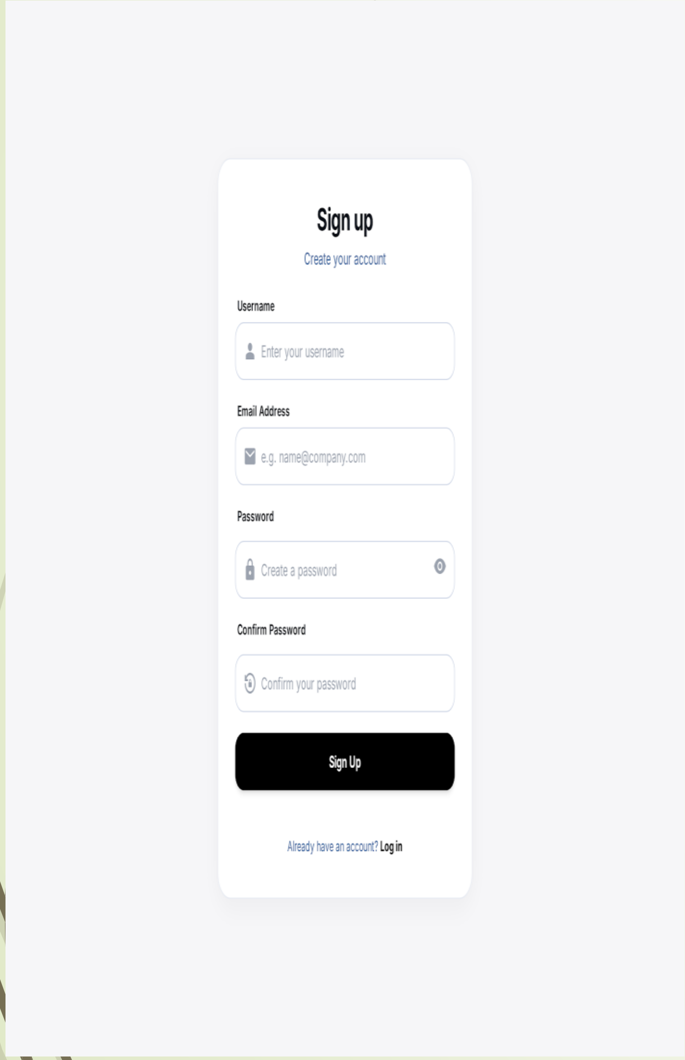
Entity Relationship Diagram

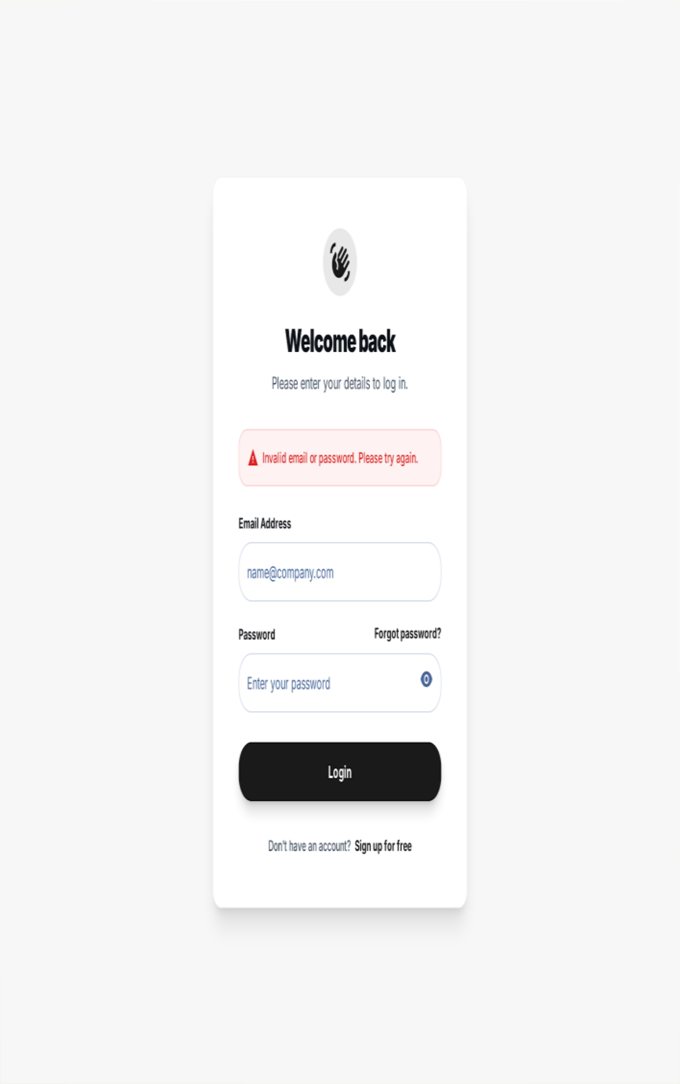


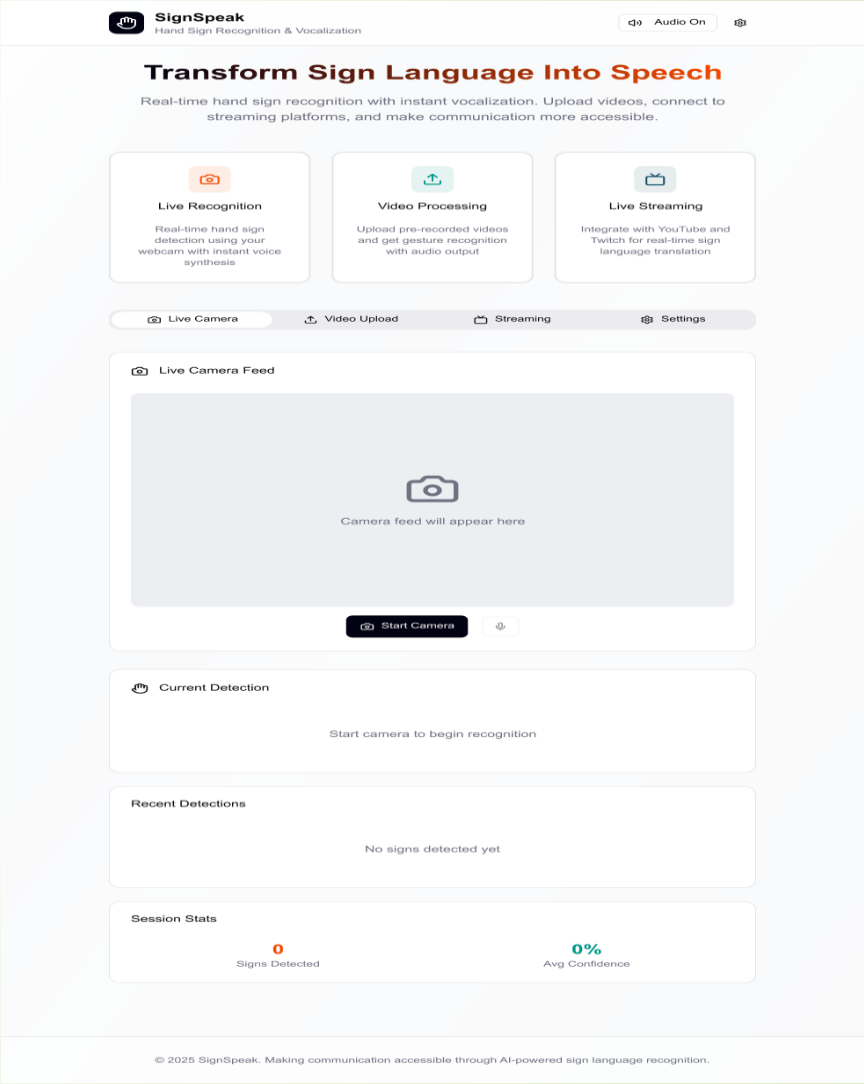
System Architecture

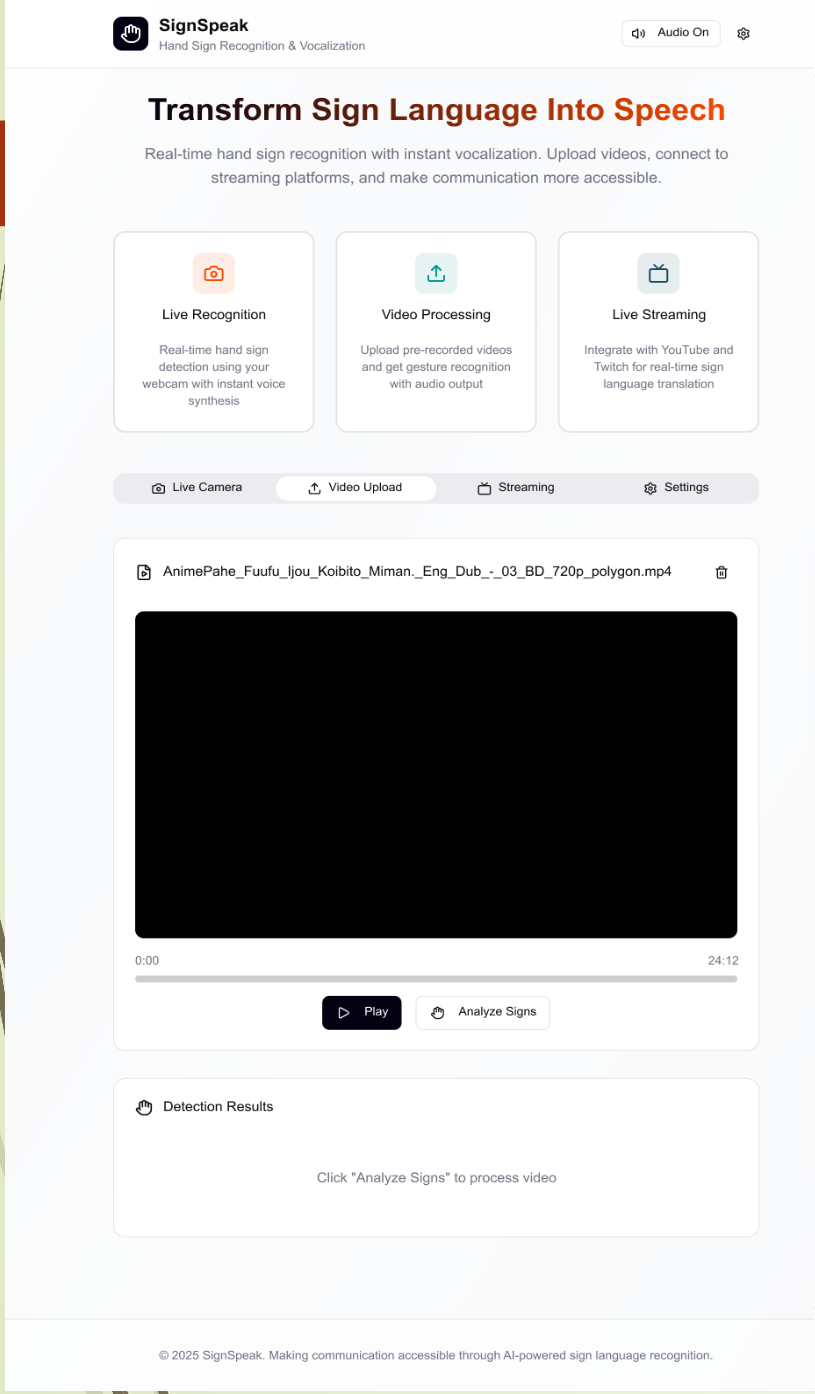


User Interface Design








SignSpeak
Hand Sign Recognition & Vocalization
Audio On
Transform Sign Language Into Speech
Real-time hand sign recognition with instant vocalization. Upload videos, connect to streaming platforms, and make communication more accessible.
Live Recognition
Real-time hand sign detection using your webcam with instant voice synthesis
Video Processing
Upload pre-recorded videos and get gesture recognition with audio output
Live Streaming
Integrate with YouTube and Twitch for real-time sign language translation
Live Camera
Video Upload
Streaming
Settings
AnimePahe_Fuufu_Ijou_Koibito_Miman._Eng_Dub_-_03_BD_720p_polygon.mp4
0:00
24:12
Play
Analyze Signs
Detection Results
Click "Analyze Signs" to process video
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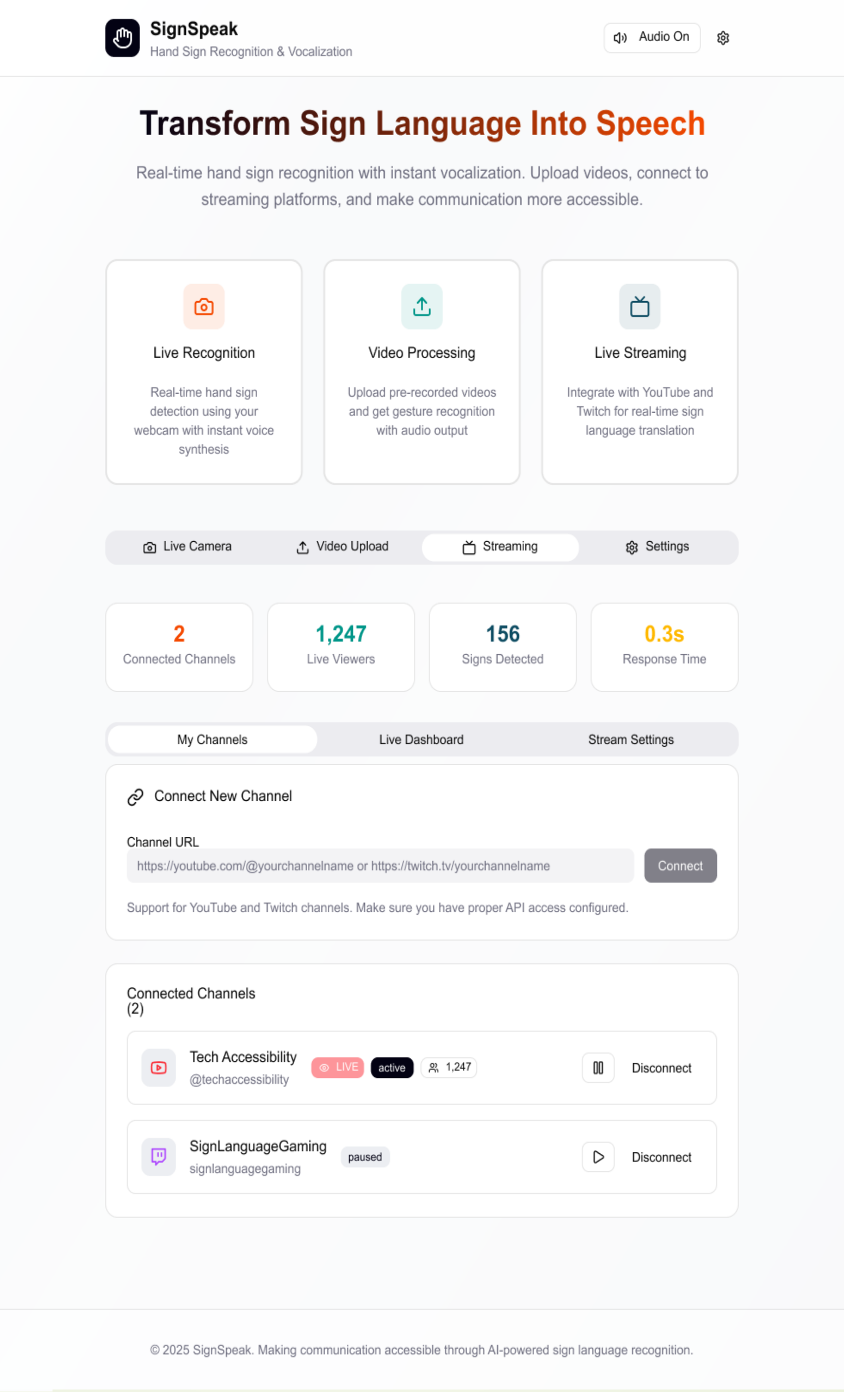
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Live Camera
Video Upload
Streaming
Settings
2
Connected Channels
1,247
Live Viewers
156
Signs Detected
0.3s
Response Time
My Channels
Live Dashboard
Stream Settings
Connect New Channel
Channel URL
https://youtube.com/@yourchannelname or https://twitch.tv/yourchannelname
Connect
Support for YouTube and Twitch channels. Make sure you have proper API access configured.
Connected Channels (2)
Tech Accessibility
@techaccessibility
LIVE
active
1,247
Disconnect
SignLanguageGaming
signlanguagegaming
paused
Disconnect
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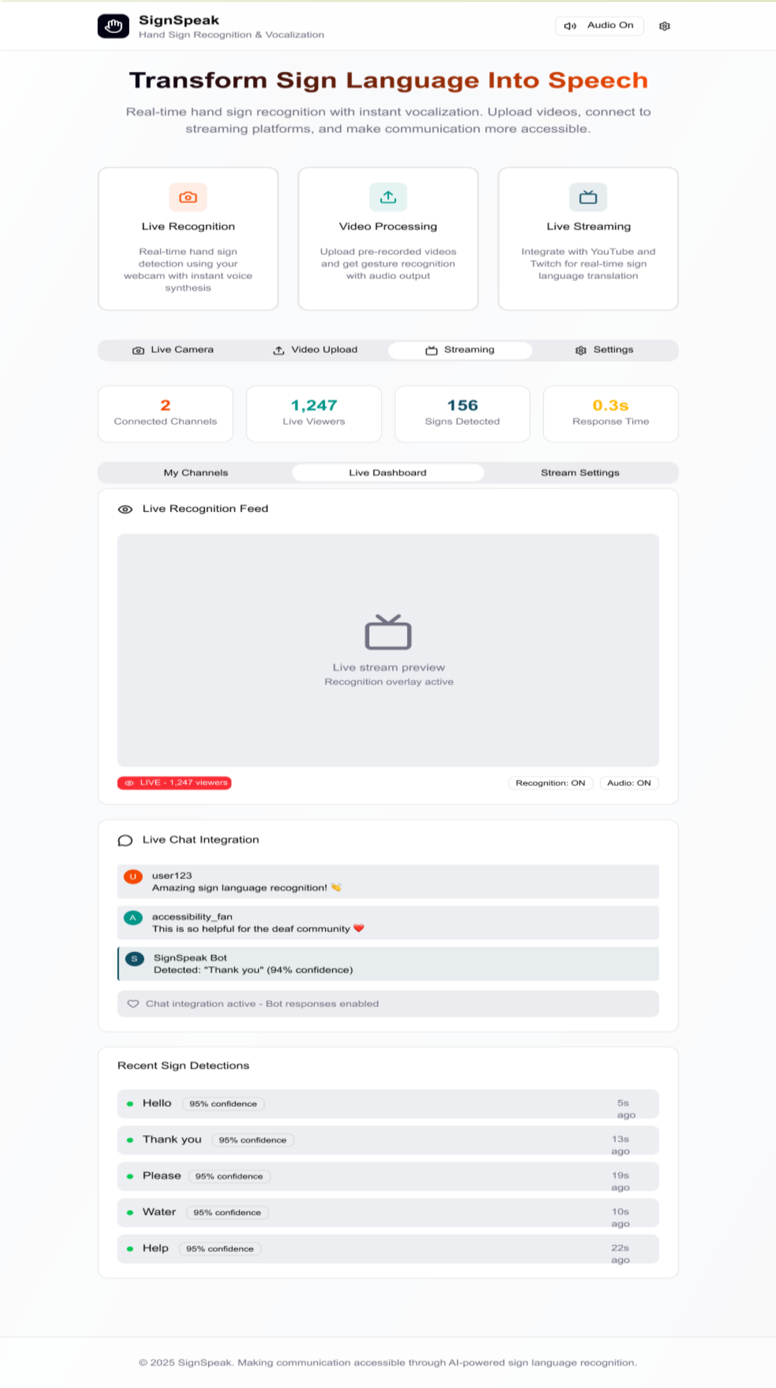
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My Channels
Live Dashboard
Stream Settings
Live Recognition Feed
Live stream preview
Recognition overlay active
LIVE • 1,247 viewers
Recognition: ON
Audio: ON
Live Chat Integration
user123
Amazing sign language recognition! 👏
accessibility_fan
This is so helpful for the deaf community ❤️
SignSpeak Bot
Detected: "Thank you" (94% confidence)
Chat integration active - Bot responses enabled
Recent Sign Detections
Hello
95% confidence
5s ago
Thank you
95% confidence
13s ago
Please
95% confidence
19s ago
Water
95% confidence
10s ago
Help
95% confidence
22s ago
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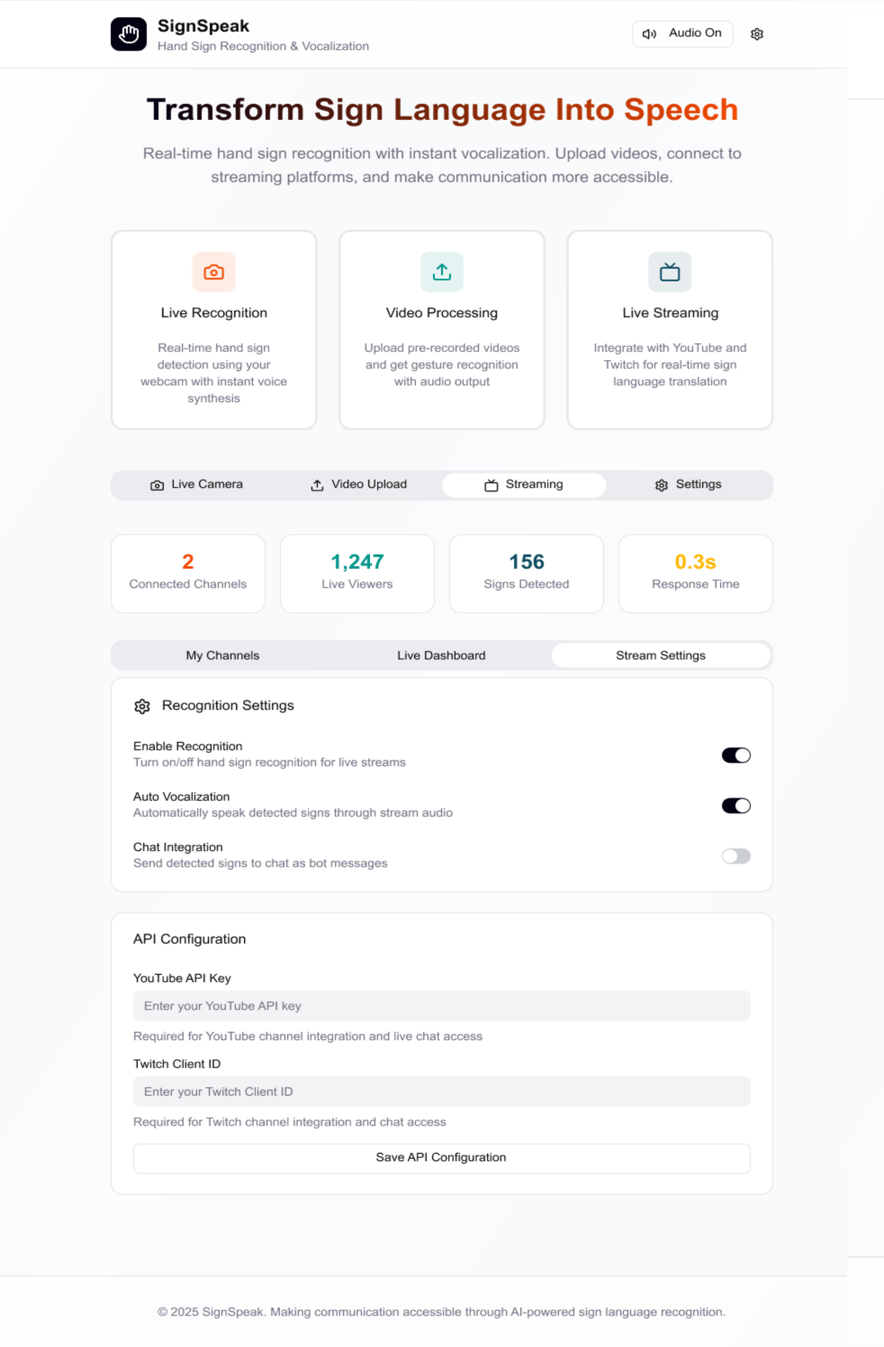
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2
Connected Channels
1,247
Live Viewers
156
Signs Detected
0.3s
Response Time
My Channels
Live Dashboard
Stream Settings
Recognition Settings
Enable Recognition
Turn on/off hand sign recognition for live streams
Auto Vocalization
Automatically speak detected signs through stream audio
Chat Integration
Send detected signs to chat as bot messages
API Configuration
YouTube API Key
Enter your YouTube API key
Required for YouTube channel integration and live chat access
Twitch Client ID
Enter your Twitch Client ID
Required for Twitch channel integration and chat access
Save API Configuration
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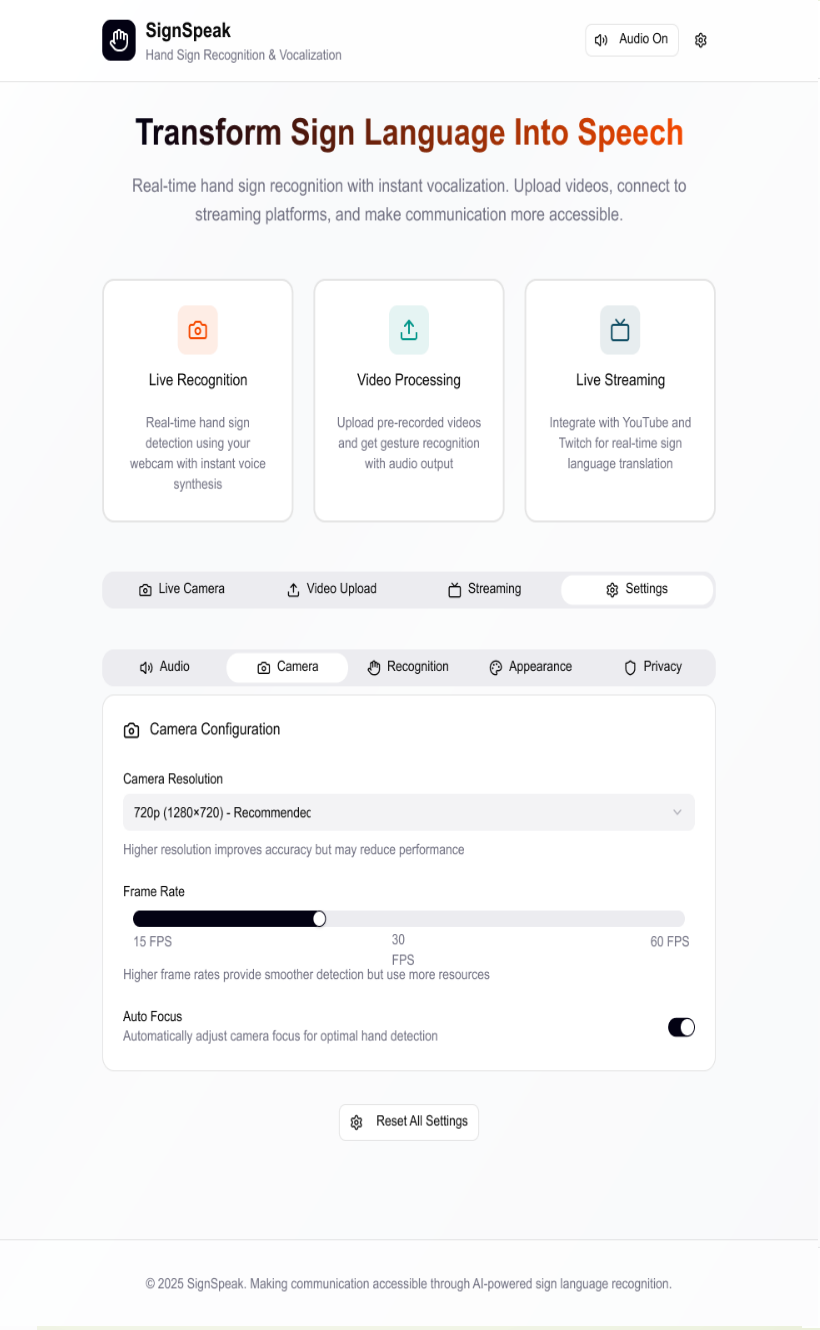
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Live Camera
Video Upload
Streaming
Settings
Audio
Camera
Recognition
Appearance
Privacy
Camera Configuration
Camera Resolution
720p (1280×720) - Recommendec
Higher resolution improves accuracy but may reduce performance
Frame Rate
15 FPS
30 FPS
60 FPS
Higher frame rates provide smoother detection but use more resources
Auto Focus
Automatically adjust camera focus for optimal hand detection
Reset All Settings
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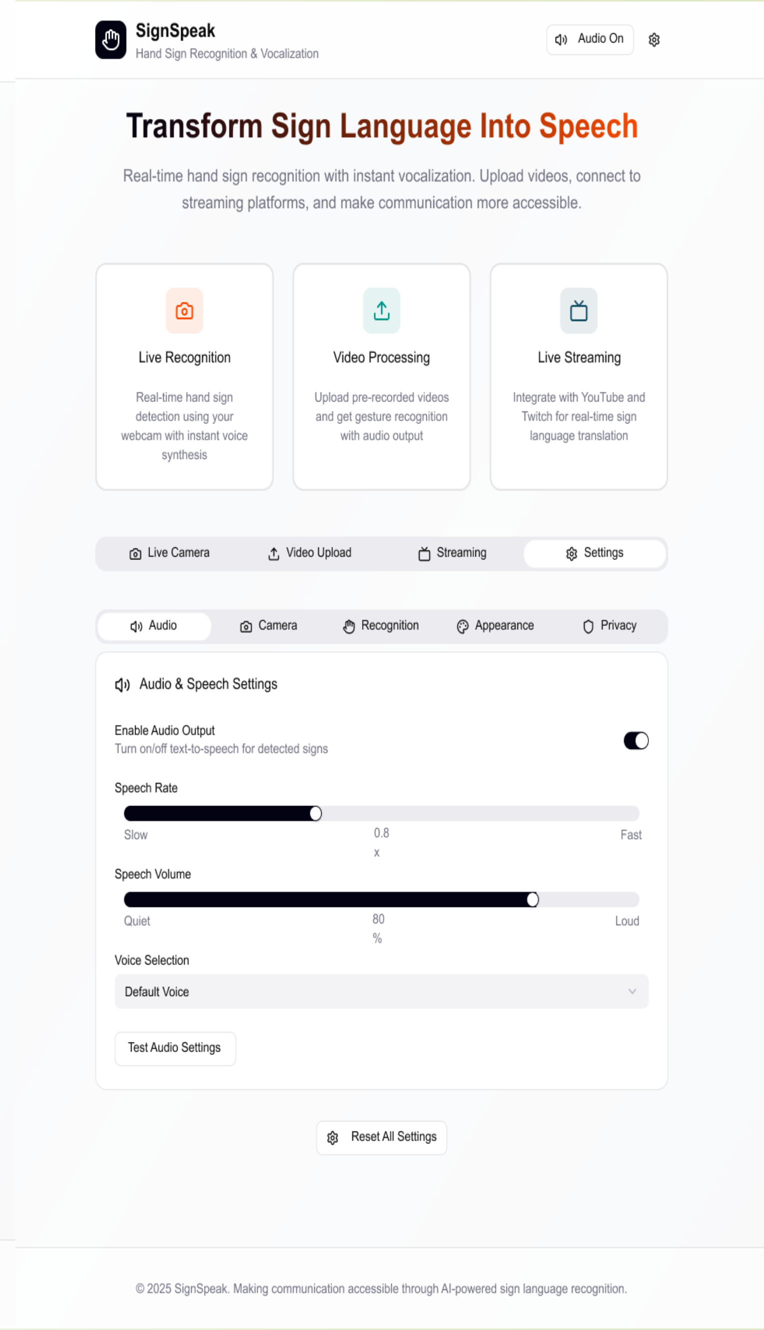
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Streaming
Settings
Audio
Camera
Recognition
Appearance
Privacy
Audio & Speech Settings
Enable Audio Output
Turn on/off text-to-speech for detected signs
Speech Rate
Slow
0.8 x
Fast
Speech Volume
Quiet
80 %
Loud
Voice Selection
Default Voice
Test Audio Settings
Reset All Settings
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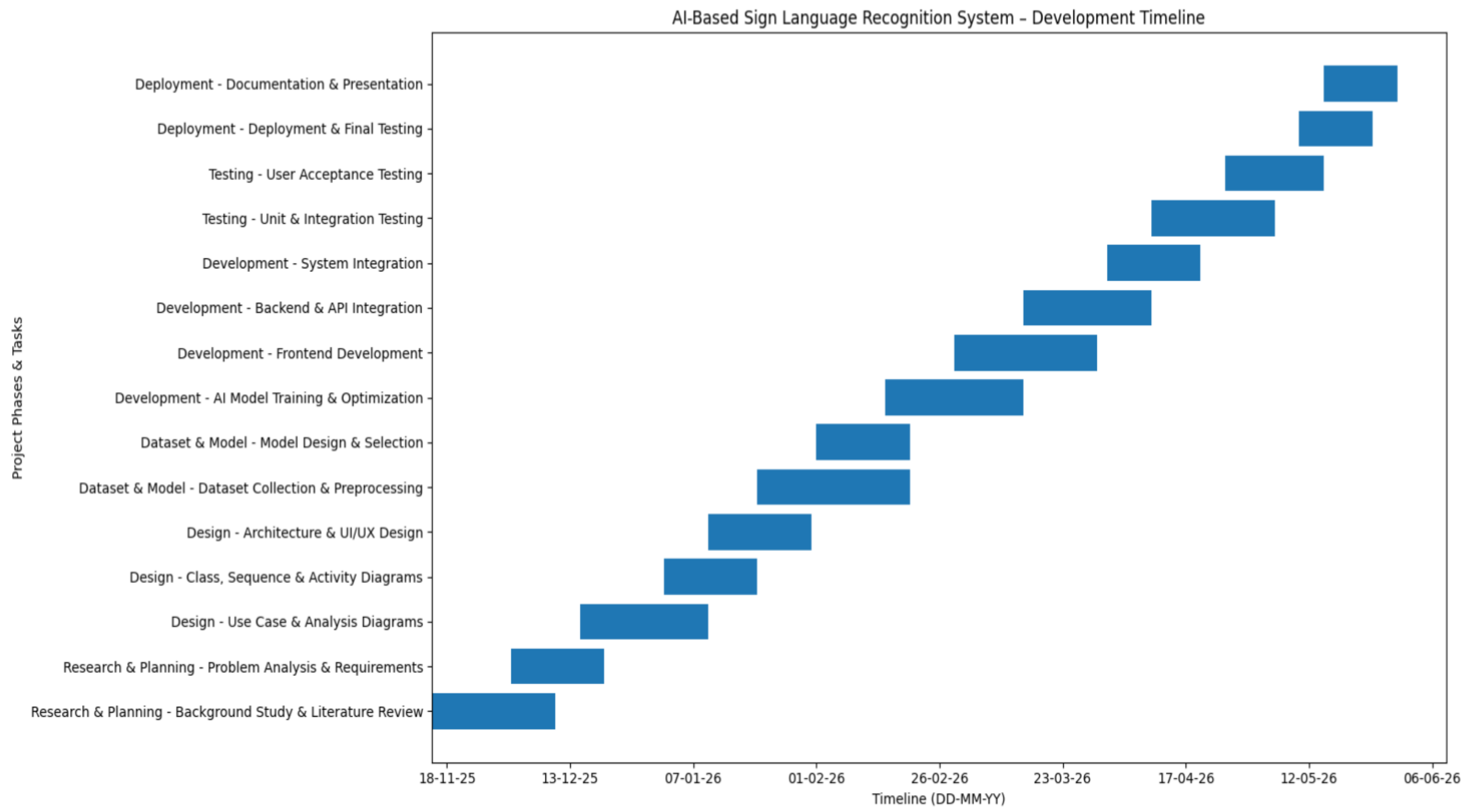
Gantt Chart


AI-Based Sign Language Recognition System – Development Timeline
Deployment - Documentation & Presentation
Deployment - Deployment & Final Testing
Testing - User Acceptance Testing
Testing - Unit & Integration Testing
Development - System Integration
Development - Backend & API Integration
Development - Frontend Development
Development - AI Model Training & Optimization
Dataset & Model - Model Design & Selection
Dataset & Model - Dataset Collection & Preprocessing
Design - Architecture & UI/UX Design
Design - Class, Sequence & Activity Diagrams
Design - Use Case & Analysis Diagrams
Research & Planning - Problem Analysis & Requirements
Research & Planning - Background Study & Literature Review
Project Phases & Tasks
18-11-25
13-12-25
07-01-26
01-02-26
26-02-26
23-03-26
17-04-26
12-05-26
06-06-26
Timeline (DD-MM-YY)

Conclusion

- In conclusion, the project provides an affordable and inclusive AI-based solution that improves communication and promotes accessibility for the hearing-impaired and non-verbal community.

THANK YOU